# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME:** **Artificial Intelligence** **COURSE CODE: CSA17**

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**COURSE OUTCOME COVERED**

***CO1:*** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*( BL3)*

*Assessment Tool Weightage: 50%*

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**QUESTIONS: 2**

**Duration : 1 Hour**
**Total Marks:30**

Annexure A

**<u>Constraint-Based Problem Solving</u>**

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***Code of Conduct:***

*I* P.Karthik 192524888 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** Karthik

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1. A college must assign 4 employees (E1, E2, E3, E4) to 4 administrative tasks (Admissions, Accounts, Records, IT Support) such that:
i. E1 can only handle Admissions or Accounts
ii. E2 must be assigned before E3 (Admissions < Accounts < Records < IT Support)
iii. Only one employee per task
iv. E4 cannot handle Records
v. All employees must be assigned exactly one task

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule.

2. A security patrol robot operates in a 6x6 building-floor grid from Start (S) to a checkpoint (G). Certain zones are restricted (no-entry) and some corridors carry extra monitoring cost.
i. Movement allowed: Up, Down, Left, Right
ii. Normal move cost = 1, monitored-corridor move cost = 2
iii. Cannot pass through restricted (no-entry) cells
iv. The robot must minimize total path cost

Using Uniform Cost Search (UCS), show step-by-step node expansion and priority-queue updates, and determine the optimal path and cost.

# Apply Backtracking search to find

## Given:

Employees: $E_1, E_2, E_3, E_4$

Tasks: Addmissions (A)
Accounts (AC)
Records (R)
IT Support (IT)

Constraints:

1) $E_1 \rightarrow$ Admissions or Accounts only
2) E2 must be assigned before E3.
   - order: Admissions < Accounts < Records < IT
3) Only one employee per task
4) E4 cannot handle Records

Step 1:

Assign E1 $\rightarrow$ Admissions
Current Assignment.

Step 2:

Assign E2 $\rightarrow$ Accounts
Reason: Accounts comes before Records and IT

Step 3:

Assign E3 $\rightarrow$ Records
- E2 is assigned to Accounts, which comes before Records.

Step 4:

Assign E4 $\rightarrow$ IT support.
- Records is not assigned to E4.

## Final Valid schedule

| Employee | task |
|---|---|
| $E_1$ | Addmissions |
| $E_2$ | Accounts |
| $E_3$ | Records |
| $E_4$ | IT support. |

2) **Uniform cost search:**

Assume the following 6X6 Grid

| | | | | | |
|---|---|---|---|---|---|
| S | 1 | 1 | X | 2 | 1 |
| 1 | X | 1 | 1 | 2 | 1 |
| 1 | 1 | 2 | X | 1 | 1 |
| X | 1 | 1 | 1 | 2 | 1 |
| 1 | 2 | X | 1 | 1 | 1 |
| 1 | 1 | 1 | 2 | 1 | ~~X~~G |

- S = start
- G = Goal
- X = Restricted
- 1 = Normal cost.
- 2 = Monitored corridor

Optimal path:

S<br>
↓<br>
↓<br>
→<br>
→<br>
↓<br>
→<br>
→<br>
↓<br>
↓<br>
G

## ost calculation:

| Move | cost |
|---|---|
| S→1 | 1 |
| Down | 1 |
| Right | 1 |
| Right | 2 |
| Down | 1 |
| Right | 1 |
| Down | 2 |
| Down | 1 |

Total cost = 10

Minimum cost = 10

## Priority Queue Table:

| Step | Expanded node | Priority Queue. |
|---|---|---|
| 1 | S | (S,0) |
| 2 | (0,1) | (1,0,1), (0,2,2) |
| 3 | (1,0) | (0,2,2), (2,0,2) |
| 4 | (0,2) | (2,0,2) (1,2,3) |
| 5 | Continue | lowest-cost node first |
| Final | Goal | Cost = 10 |

## Optimal path:

S → (1,0) → (2,0) → (2,1) → (2,2) → (3,2) → (3,3) → (4,3) → (5,3) → G